

VPA IB DP CS — Topic 1: System Fundamentals (Weeks 1 – 9)

Topic 1: System Fundamentals 系统基础 Xìtǒng Jīchǔ

IB Syllabus: Topic 1 (1.1 – 1.2) | Level: SL + HL | Duration: 9 weeks (40 min × 5 lessons/week = 30 hours)

Topic 1.1: Systems in Organizations 组织中的系统 Zǔzhī zhōng de Xìtǒng

Week 1 — Introduction to Systems 系统简介 Xìtǒng Jiǎnjiè

Lesson	Content	Activities	Differentiation
1	What is a system? Open vs closed systems. Systems in everyday life.	Brainstorm: systems at VPA (timetable, canteen, library)	Bronze: Label system diagrams; Silver: Describe a chosen system; Gold: Analyse system boundaries
2	Components of an information system: hardware, software, data, procedures, people (硬件 yìngjiàn, 软件 ruǎnjiàn, 数据 shùjù)	Match components to descriptions (bilingual cards)	Bronze: Matching; Silver: Give examples; Gold: Evaluate missing component impact
3	Roles in system development: Systems analyst, project manager, end-user, client (系统分析师 xìtǒng fēnxī shī)	Role-play: interview a "client" about their needs	All: Participate in role-play with differentiated question scripts
4	System life cycle models: Waterfall, Agile, RAD (瀑布模型 pùbù móxíng, 敏捷开发 mǐnjié kāifā)	Compare models using a Venn diagram	Bronze: Fill Venn diagram; Silver: Explain one model; Gold: Evaluate which model suits a given scenario
5	TOK: Can we ever fully understand a client's needs? Formative: Exit ticket on system components	Discussion + written reflection	All: Contribute to discussion; Gold: Link to knowledge questions

TOK Knowledge Question: When a systems analyst gathers requirements, they construct a representation of reality. To what extent is this representation an accurate map of the client's world, and what ethical implications arise when the map diverges from the territory?

Week 2 — Planning and System Installation 规划与系统安装 Guīhuà yǔ Xìtǒng Ānzhuāng

Lesson	Content	Activities
1	Feasibility study: technical, economic, operational feasibility (可行性研究 kěxíngxìng yánjiū)	Case study: Should VPA adopt a new LMS? Students assess feasibility dimensions
2	Project planning: Gantt charts, milestones, critical path (甘特图 gāntè tú)	Create a Gantt chart for a system replacement project using spreadsheet software

Lesson	Content	Activities
3	Changeover methods: direct, parallel, phased, pilot (切换方法 qiēhuàn fāngfǎ)	Scenario cards: match method to situation; justify choice using IB command terms
4	Data migration strategies and risks (数据迁移 shùjù qiānyí)	Plan a data migration for a school records system
5	Formative Assessment: Short-answer test on planning and changeover (20 marks, Paper 1 style)	Individual test + peer marking with markscheme

Week 3 — User Focus and Change Management 用户关注与变革管理 Yònghù Guānzhù yǔ Biàngé Guǎnlǐ

Lesson	Content	Activities
1	User-centered design principles (以用户为中心的设计 yǐ yònghù wéi zhōngxīn de shèjì)	Heuristic evaluation of a familiar app/website
2	User training strategies: workshops, manuals, e-learning, help desks	Design a training programme for VPA staff on a new system
3	Managing change: resistance, communication, stakeholder engagement	Role-play: address resistant staff member; use persuasive techniques
4	Post-implementation review: evaluating success against objectives	Write a post-implementation review for a case study system
5	Consolidation: Topic 1.1 revision carousel	Station rotation with differentiated tasks

Topic 1.2: System Design Basics 系统设计基础 Xìtǒng Shèjì Jīchǔ

Week 4 — Modeling Systems: ER Diagrams 实体关系图 ShíTǐ Guānxi Tú

Lesson	Content	Activities	Differentiation
1	Introduction to entity-relationship modeling: entities, attributes, relationships (实体 shíTǐ, 属性 shǔxìng, 关系 guānxi)	Identify entities in a school library system	Bronze: Circle entities in description; Silver: List attributes; Gold: Define relationships with cardinality
2	Cardinality and participation: 1:1, 1:N, M:N (基数 jīshù)	Match relationship types to scenarios	Bronze: Match given; Silver: Determine from description; Gold: Resolve M:N into associative entities
3	Constructing ER diagrams from written descriptions	Guided practice: VPA library system ER diagram	Scaffolded worksheets with decreasing support
4	Constructing ER diagrams independently	Independent: Design ER for VPA cafeteria ordering system	Bronze: 3 entities minimum; Silver: 4+ with correct cardinality; Gold: Normalisation to 3NF discussed

Lesson	Content	Activities	Differentiation
5	TOK: Do abstractions like ER diagrams capture the truth of a system, or do they impose a particular way of seeing?	Socratic seminar	All participate; Gold facilitates

TOK Knowledge Question: ER diagrams are simplifications of complex realities. When we reduce a real-world situation to entities and relationships, what knowledge do we gain, and what do we inevitably lose? Is a model that omits detail more or less true than the reality it represents?

Week 5 — Flowcharts and IPO Tables 流程图与IPO表 Liúchéng Tú yǔ IPO Biǎo

Lesson	Content	Activities
1	Flowchart symbols: start/end, process, decision, input/output, connector (流程图 liúchéngtú)	Symbol bingo + trace a given flowchart
2	Constructing flowcharts from algorithms	Paired: convert pseudocode to flowchart (temperature converter)
3	IPO tables: Input, Processing, Output (输入/处理/输出 shūrù/chǔlǐ/shūchū)	Create IPO tables for 3 standard problems
4	System flowcharts vs program flowcharts	Compare and contrast; identify appropriate use cases
5	Formative: 30-minute assessment — ER diagram + flowchart construction from scenario	Individual; peer-mark with criteria

Week 6 — Data Flow Diagrams (HL Extension) 数据流图 (HL) Shùjù Liú Tú

HL Only Content: Context diagrams, Level 0 DFDs, data stores, processes, external entities. Balancing DFD levels.

Lesson	Content	Activities
1	Context diagrams: system boundary, external entities (上下文图 shàngxiàwén tú)	Draw context diagram for VPA attendance system
2	Level 0 DFD: processes, data stores, data flows	Decompose context diagram to Level 0
3	Balancing DFDs: ensuring consistency across levels	Validate a given set of DFDs; identify errors
4	Common DFD errors and how to avoid them	Error-spotting exercise + correction
5	HL Consolidation: Complete system design package (ER + DFD + flowchart) for case study	Extended task with marking rubric

Week 7 — System Design & Review

Lesson	Content	Activities
1	User interface design principles: consistency, feedback, error prevention, accessibility	Critique UI examples; redesign a poor interface

Lesson	Content	Activities
2	Documentation: technical documentation vs user documentation	Write user guide excerpt for a given system
3	System testing: unit, integration, system, acceptance testing (单元测试 dānyuán cèshì, 集成测试 jíchéng cèshì)	Match test type to purpose; plan test schedule
4	Revision: Topic 1.1 and 1.2 concept map building	Collaborative concept map on whiteboard / digital tool
5	Revision: Exam technique — how to answer 4-mark, 8-mark, and 12-mark questions	Practice questions with timed responses

Week 8 — Topic 1 Consolidation & IA Foundation

Lesson	Content	Activities
1	Case study analysis: a real-world system implementation (success and failure)	Group analysis using structured framework
2	Case study analysis continued	Present findings; peer questioning
3	IA Introduction: What is the IA? Real client, real problem. Criteria overview.	Review exemplar IAs; identify client and problem
4	IA: Brainstorming client contacts and problem areas in school community	Mind map; initial client list
5	Formative: Topic 1 progress test (Paper 1 style, 40 marks)	Timed conditions; self-assessment against markscheme

Week 9 — Topic 1 End-of-Topic Assessment

Lesson	Content	Activities
1	Revision: Key concepts and common misconceptions	Quiz; error analysis
2	Revision: Exam practice — 12-mark essay planning	Plan 3 essays in groups; share structures
3	Assessment: Topic 1 End-of-Topic Test (1 hour, Paper 1 style)	Formal test conditions
4	Test feedback and target setting	Review scripts; set SMART targets for Topic 2
5	Introduction to Topic 2: What is inside a computer?	Disassembly video; label a computer diagram

Key Vocabulary (Topic 1)

English	中文	Pinyin	Definition
System	系统	xìtǒng	A set of interconnected components working together to achieve a purpose

English	中文	Pinyin	Definition
Feasibility study	可行性研究	kěxíngxìng yánjiū	Analysis of technical, economic, and operational viability
Changeover	系统切换	xìtǒng qiēhuàn	The process of replacing an old system with a new one
Entity	实体	shí tǐ	A distinct object or concept about which data is stored
Relationship	关系	guānxì	An association between two or more entities
Cardinality	基数	jīshù	The number of instances in one entity related to instances in another
Flowchart	流程图	liúchéngtú	A diagram representing the sequence of operations in a process
IPO table	输入处理输出表	shūrù chǔlǐ shūchū biǎo	A table listing inputs, processing steps, and outputs for a system
Stakeholder	利益相关者	lìyì xiāngguān zhě	Anyone with an interest in or affected by a system
Milestone	里程碑	lǐchéngbēi	A significant point or event in a project timeline

Teacher Checklist — End of Topic 1:

- ☐ All IB 1.1 and 1.2 syllabus points covered
- ☐ TOK connections documented and discussed
- ☐ Formative assessments marked with feedback
- ☐ EAL vocabulary recorded in student glossaries
- ☐ IA client brainstorming completed
- ☐ Bronze/Silver/Gold tracking updated
- ☐ Victoria Park Academy Shenzhen | IB DP Computer Science | Topic 1 Scheme of Work